CORE VALUES | QUALITY PRODUCTS | SUPERIOR SERVICE

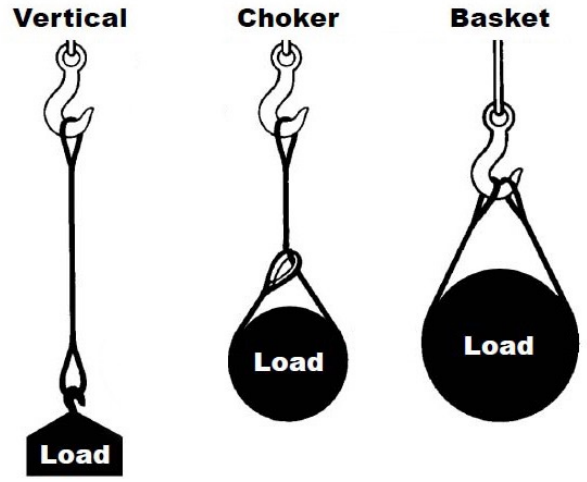
CORE® LIFTING PRODUCTS

CORE-TEX®

WEB SLINGS

CoreLifting.com

General Information
Basic Hitches



VERTICAL, or straight, attachment is simply using a sling to connect a lifting hook to a load. Full rated lifting capacity of the sling may be utilized but must not be exceeded. A tagline should be used to prevent load rotation, which may damage a sling. When two or more slings are attached to the same lifting hook, the total hitch becomes, in effect, a lifting bridle, and the load is distributed equally among the individual slings.

CHOKER hitches reduce lifting capability of a sling since this method of rigging affects ability of the wire rope components to adjust during the lift. A choker is used when the load will not be seriously damaged by the sling body — or the sling damaged by the load, and when the lift requires the sling to snug up against the load.

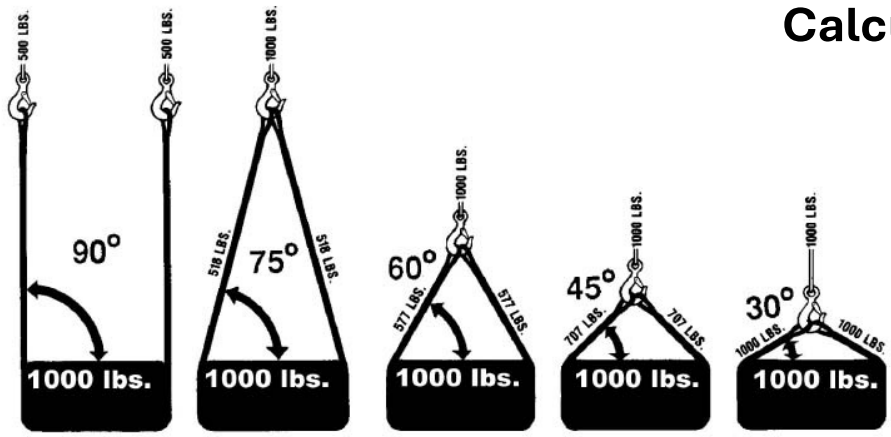
The diameter of the bend where the sling contacts the load should keep the point of choke against the sling BODY — never against a splice or the base of the eye. When a choke is used at an angle of less than 120 degrees (see next page), the sling-rated capacity must be adjusted downward.

A choker hitch should be pulled tight before a lift is made — NOT PULLED DOWN DURING THE LIFT. It is also dangerous to use only one choker hitch to lift a load which might shift or slide out of the choke.

BASKET hitches distribute a load equally between the two legs of a sling ... within limitations described above.

General Information
Load Calculating

Calculating the Load on Each Leg of a Sling



As the included angle between the legs of a sling decreases, the load on each leg increases. The effect is the same whether a single sling is used as a basket or two slings are used with each in a straight pull, as with a 2-legged bridle.

Anytime pull is exerted at an angle on a leg—or legs—of a sling, the load per leg can be determined by using the data in the table above. Proceed as follows to calculate this load—and determine the rated capacity required of the sling, or slings, needed for a lift.

1.First, divide the total load to be lifted by the number of legs to be used. This provides the load per leg if the lift were being made with all legs being vertically.

2.Determine the angle between the legs of the sling and the horizontal.

3.Then MULTIPLY the load per leg (as computed above) by the Load Factor for the leg angle being used (from the table at the bottom) – to compute the ACTUAL LOAD on each leg for this lift and angle. THE ACTUAL LOAD MUST NOT EXCEED THE RATED SLING CAPACITY.

Thus, in the above drawing (sling angle at 60°): 1000 ÷ 2 = 500 (Load Per Leg if a vertical lift) 500 x 1.154 = 577 lbs. – ACTUAL LOAD on each leg at the 60° included angle being used.

In the above drawing (sling angle of 45°): 1000 ÷ 2 = 500 (Load Per Leg if a vertical lift) 500 x 1.414 = 707 lbs. = ACTUAL LOAD on each leg at the 45° horizontal angle being used.

General Information
Bridle Legs – Angle Effects

Angles of Bridles

The horizontal angle of bridles with 3 or more legs is measured the same as the horizontal sling angle of 2-legged hitches. In this case, where a bridle designed with different leg lengths results in horizontal angles, the leg with the smallest horizontal angle will carry the greatest load. Therefore, the smallest horizontal angle is used in calculating actual leg load and evaluating the rated capacity of the sling proposed.

Leg Angle (Degrees)	Load Factor
90	1.000
85	1.003
80	1.015
75	1.035
70	1.064
65	1.103
60	1.154
55	1.220
50	1.305
45	1.414
40	1.555
35	1.743
30	2.000

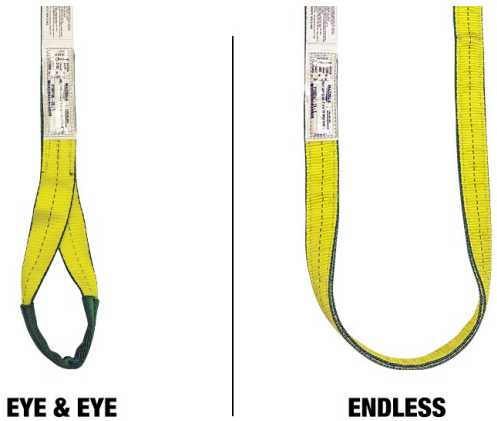
General Information
Web Sling Configurations

Web slings are pretty similar in configuration to other sling types, like wire rope slings.

Web slings come in several different configurations, including:

- Single or multi-leg bridle assemblies
- Eyes formed on both ends of the sling
- An endless or infinite loop configuration (similar to a synthetic roundsling)
- With fittings on both ends of the sling.

Hardware manufacturers make special hooks for web slings, and instead of your traditional eye on a hook, they actually make a flat surface on the hook. This way, the web sling will sit properly against the hook without getting cinched up, which allows you to get the maximum lifting strength.



General Information
Web Sling Types

Type I Triangle/Choker - (TC)
Web sling made with a triangle fitting on one end and slotted triangle choker fitting on the other end. It can be used in a vertical, basket or choker hitch.

Type II Triangle x Triangle - (TT)
Web sling made with a triangle fitting on both ends. It can be used in a vertical or basket hitch only.

Type III Eye x Eye - (EE)
Web sling made with a flat loop eye on each end with loop eye opening on same plane as sling body. This type of sling is sometimes called a flat eye x eye or double eye sling.

Type IV Reversed Eye - (RE)
Web sling made with both loop eyes formed as in Type III, except that the loop eyes are turned to form a loop eye, which is at a right angle to the plane of the sling body. This type of sling is commonly referred to as a twisted eye sling.

Type V Endless Type - (EN)
Endless web sling, sometimes referred to as a grommet. It is a continuous loop formed by joining the ends of a piece of webbing together with a load bearing splice.

General Information
Measuring Sling Length

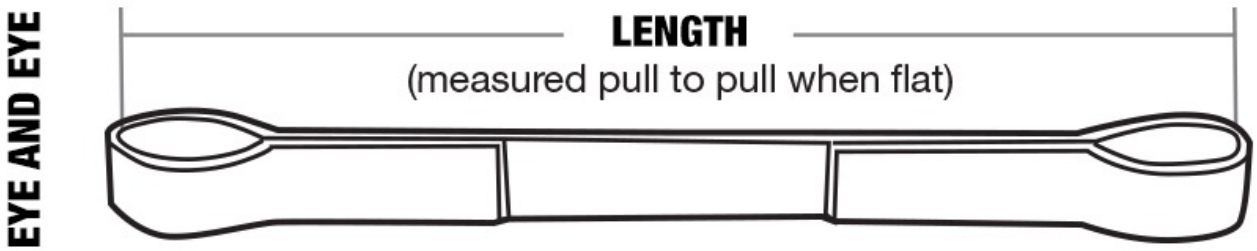
How to accurately measure a web sling.

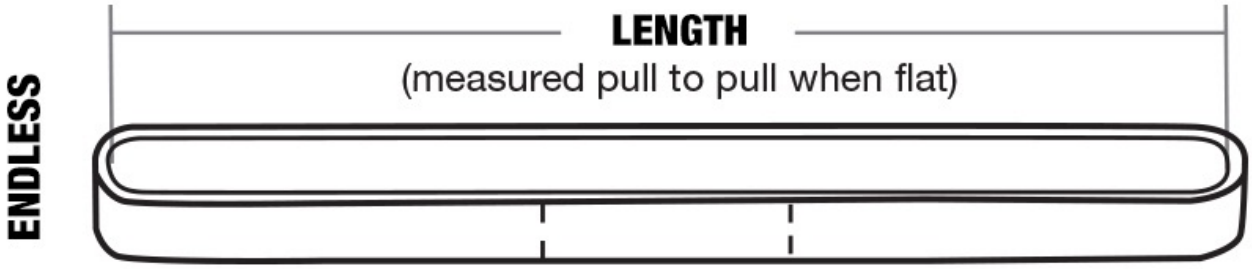
Synthetic web slings can come in various lengths. Many fabricators have standardized product offerings, including incremental increases in length ranging anywhere from 2’ to 20’. However, it’s also common for fabricators to offer customized sling lengths, so you can get a web sling designed to your application’s exact specifications.

When determining sling length, measure from one load-bearing point to the opposite load-bearing point.

An eye and eye sling length is measured from the inside edge of the eyes.

Multi-leg sling assemblies, or single-leg slings with fittings, are measured from the load-bearing point of rings, hooks, or fittings.





General Information
Polyester vs Nylon

Polyester Slings and Nylon Slings

For many years, most of the world outside of the United States used Polyester slings, while the U.S. used Nylon. Though nearly identical in strength, there are differences in how each reacts to loading and they're interaction with chemicals. In the past 10 years, the U.S. market has largely migrated to Polyester, and this is the primary webbing fiber used in Core-Tex web slings.

Polyester Web Slings

- Approximately 3% stretch at rated capacity for more load control.
- Polyester is a slightly softer fiber and less likely to scratch surfaces of what is being lifted.
- Absorbs less liquid than Nylon.
- Is non-conductive.
- Resistant to acidic environments and bleaching agents.
- Is less costly than Nylon.
- Not recommended for alkaline environments including aldehydes, ethers, and strong alkalis.
- Cannot be used in temperatures greater than 194F or below -40F.

Nylon Web Slings

- 8-10% stretch at rated capacity. This helps absorb shock loads, but this must be accounted for, especially in low headroom situations.
- Unaffected by grease and oil.
- Resistant to alkalis such as aldehydes, ethers and strong alkalis.
- Not recommended for acidic environments or exposure to bleaching agents.
- Absorbs liquid which can add to or cause more stretch under load.
- Will conduct electricity.
- Cannot be used in temperatures greater than 194F or below -40F.

General Information
Plies

Number of Plies

The ply number, or number of plies, refers to the layers of webbing that make up the body of the sling. A web sling is generally available in 1, 2, 3, or 4-ply configurations. Adding more plies to the sling provides added strength to the body of the sling.

Lifting capacity is going to determine how many plies are needed to execute your lifts safely. More times than not, when you have higher-capacity lifts, you're going to need more plies on your web slings.

For example, if the breaking strength of a 1-ply web sling = X, then...

- The breaking strength of an equivalent 2-ply web sling = 2X
- The breaking strength of an equivalent 3-ply web sling = 3X
- The breaking strength of an equivalent 4-ply web sling = 4X

1- Ply

2- Ply

3- Ply

4 - Ply

General Information
Width

Web Sling Width

With a wider sling body, the load is distributed over a larger sling surface area. Wider width slings provide greater protection to delicate or finished surfaces, as well as better load balance and control for large or heavy loads.

Typical standard sling lengths can range anywhere from 1” to 12” in width. Larger sizes are available as special-orders, or in a special wide lift sling configuration.

If you don’t want to use a three-ply web sling, you may want to increase the width of your synthetic material for better load control.

A lot of times, if you’re using your sling in a basket hitch configuration, that additional width is going to help keep a load balanced more than using a thinner sling. With a thinner sling, the load might move if you’re just using one. Determining what you need to execute your lifts is key to finding the right rigging gear.

How width affects strength:

Sling webbing, both polyester and nylon, has a typical breaking strength of 9,800 lbs per inch. So, 1 inch webbing has a breaking strength of 9,800 lbs. 2 inch wide webbing has double that or 19,600 lbs of breaking strength.

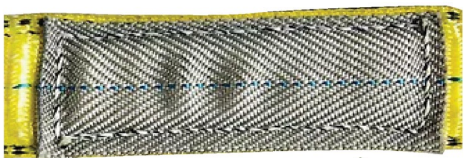
The number of plies and width of webbing combine to determine breaking strength of the sling. So, a 2 inch wide web sling that has 2 plies has a breaking strength in vertical configuration of 2 x 9,800 lbs x 2 plies or 39,200 lbs (2 x 9,800 = 19,600 lbs x 2 = 39,200 lbs.

General Information
Webbing Protection

Webbing Can Wear

Sling webbing is a fiber product vulnerable to abrasion. To guard against this, obvious wear points where the sling comes into contact with end fittings such as shackles, master links, hooks, etc., more abrasion resistant fiber webbing products such as Cordura can be sewn into the eyes of the slings.

When the sling is used in a basket configuration, Cordura type products can be wrapped over the area most vulnerable to wear and sewn onto the sling surface.



Other options are wear-sleeves that can be slid over the sling's surface.

Web Slings
Type I & Type II

CORE-TEX®

Loop Eye Table		
Sling Web Width (in.)	Recommended Minimum Loop Eye Length (in.)	Suggested Loop Eye Width (in.)
1	6	1
2	8	2
3	8	1-1/2
4	10	1-1/2
5	12	1-3/4
6	14	2
8	18	3
10	22	4
12	26	5

Type TC	Type TT **	Rated Capacities (lbs.)		
		Vertical	Choker	Basket
TC1-802	TT1-802	3,200	2,400	6,400
TC1-803	TT1-803	4,800	3,600	9,600
TC1-804	TT1-804	6,400	4,800	12,800
TC1-805	TT1-805	8,000	6,000	16,000
TC1-806	TT1-806	9,600	7,200	19,200
TC1-808	TT1-808	12,800	9,600	25,600
TC1-810	TT1-810	16,000	12,000	32,000
TC1-812	TT1-812	19,200	14,400	38,400
TC1-816	TT1-816	25,500	19,200	51,000
TC1-818	TT1-818	28,700	21,600	57,400
TC1-820	TT1-820	32,000	24,000	64,000
TC1-824	TT1-824	38,400	28,800	76,800
TC2-802	TT2-802	6,400	4,800	12,800
TC2-803	TT2-803	8,600	6,500	17,200
TC2-804	TT2-804	11,500	8,600	23,000
TC2-805	TT2-805	14,000	10,500	28,000
TC2-806	TT2-806	16,800	12,600	33,600
TC2-808	TT2-808	22,400	16,800	44,800
TC2-810	TT2-810	28,000	21,000	56,000
TC2-812	TT2-812	32,000	25,600	64,000

Tapering: Flat, Twisted Eye and Endless Slings are tapered at 3" and wider unless specified differently when ordering.
*** Please Note:** ***Polyester webbing is not available over 10" wide.***
Three and four-ply slings are available.
** TT can not be used in a choker hitch.

Type I Triangle / Choker Fittings (TC) Web Slings

- These types of slings are configured with a triangle and choker fitting on either end. These types of slings are typically used in a choker hitch and can be used in vertical and basket hitches as well.

Type II Triangle / Triangle Fittings (TT) Web Slings

- This type of sling is configured with a triangle fitting on each end. These types of slings are normally used in a basket hitch but can also be used in a vertical hitch. They cannot be used as a choker.

TT1 & TT2 Specifications

- Web thickness of “Heavy-Duty” web is approximately 3/16".
- Eye length of Flat Eye and Twisted Eye varies with sling width and number of webbing piles.

Web Slings
Type III & Type IV

CORE-TEX

Flat Eye or Twisted Eye	Rated Capacities (lbs.)		
	Vertical	Choker	Basket
EE1-801	1,600	1,250	3,200
EE1-802	3,200	2,560	6,400
EE1-803	4,800	3,840	9,600
EE1-804	6,400	5,120	12,800
EE1-806	9,600	7,680	19,200
EE1-808	12,800	10,240	25,600
EE1-810	16,000	12,800	32,000
EE1-812	19,200	15,360	38,400
EE2-801	3,200	2,560	6,400
EE2-802	6,400	5,120	12,800
EE2-803	9,300	7,440	18,600
EE2-804	11,500	9,200	23,000
EE2-806	16,500	13,200	33,000
EE2-808	22,750	18,200	44,500
EE2-810	28,400	22,720	56,800
EE2-812	34,100	27,280	68,200
EE3-801	4,100	3,280	8,200
EE3-802	8,300	6,640	16,600
EE3-803	12,500	10,000	25,000
EE3-804	16,000	12,800	32,000
EE3-806	23,000	18,400	46,000
EE3-808	30,700	24,560	61,400
EE3-810	36,800	29,440	73,600
EE3-812	44,000	35,200	88,000
EE4-801	6,200	4,960	12,400
EE4-802	12,400	9,920	24,800
EE4-803	17,000	13,600	34,000
EE4-804	22,000	17,600	44,000
EE4-806	33,000	26,400	66,000
EE4-808	44,000	35,200	88,000
EE4-810	55,000	44,000	110,000
EE4-812	66,000	52,800	132,000

Type III Eye x Eye Flat Eyes Web Slings

- These types of slings have flat eyes and can be made in widths 1” thru 12” and 1, 2, 3 or 4 plies.

Type IV Eye x Eye Reverse Eyes Web Slings

- These types of slings have reverse eyes and can be made in widths 1” thru 12” and 1, 2, 3 or 4 plies.

Type III & IV Specifications

- Flat and reverse eyes are tapered beginning at 3” and wider unless specified when ordering.

Web Slings
Type V

CORE-TEX

Endless	Rated Capacities (lbs.)		
	Vertical	Choker	Basket
EN1-801	3,200	2,500	6,400
EN1-802	6,400	5,000	12,800
EN1-803	8,600	6,900	17,200
EN1-804	11,500	9,200	23,000
EN1-806	16,300	13,000	32,600
EN1-808	19,200	15,400	38,400
EN1-810	22,400	17,900	44,800
EN1-812	26,900	21,500	53,800
EN2-801	6,200	4,900	12,400
EN2-802	12,200	9,800	24,400
EN2-803	16,300	13,000	32,600
EN2-804	20,700	16,500	41,400
EN2-806	28,600	23,000	57,200
EN2-808	30,700	24,500	61,400
EN2-810	33,600	26,800	67,200
EN2-812	37,600	30,000	75,200
EN3-801	8,000	6,400	16,000
EN3-802	16,000	12,800	32,000
EN3-803	21,500	17,200	43,000
EN3-804	28,700	23,000	57,400
EN3-806	40,700	32,500	81,400
EN3-808	46,000	36,800	92,000
EN3-810	51,500	41,200	103,000
EN3-812	59,200	47,300	118,400
EN4-801	10,000	8,000	20,000
EN4-802	19,800	15,800	39,600
EN4-803	26,700	21,300	53,400
EN4-804	35,600	28,400	71,200
EN4-806	50,500	40,400	101,000
EN4-808	57,600	46,000	115,200
EN4-810	67,200	53,700	134,400
EN4-812	80,700	64,500	161,400

Type V Eye x Eye Flat Eyes Web Slings

- These types of slings can be made in widths 1” thru 12” and 1, 2, 3 or 4 plies.

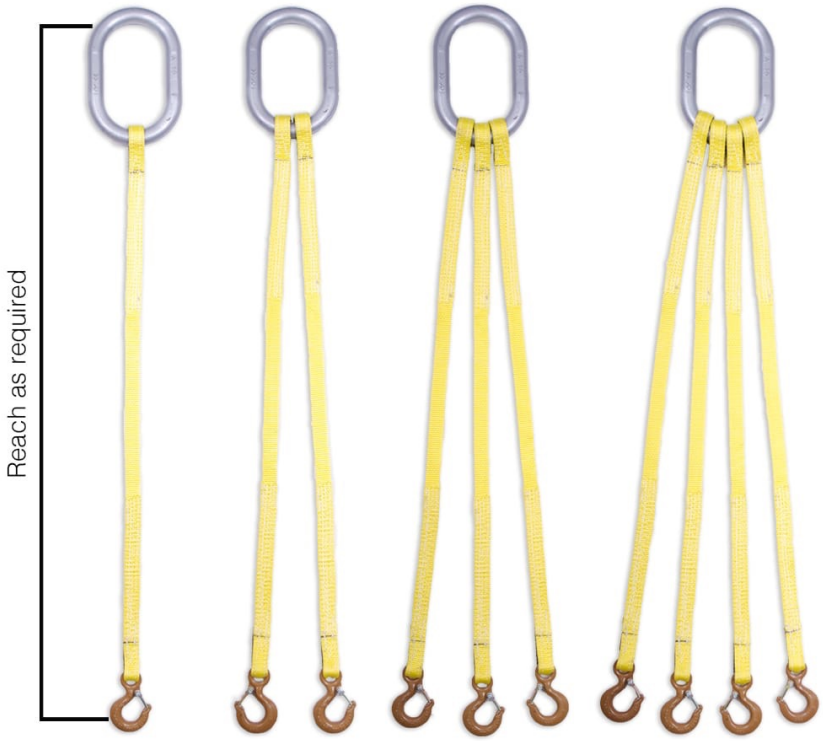
Web Slings
Multi-Leg Web Sling Bridles

CORE-TEX®

Sling Type	Webbing Width (in.)	Number of Plies	Number of Legs	Rated Capacities (lbs.)				Master Link Size (in.)	Alloy Hooks Capacity * (ton)
				Vertical	60°	45°	30°		
EE1-801	1	1	1	1600	–	–	–	1/2 x 2-1/2 x 5	1
EE1-801	1	1	2		2700	2200	1600	1/2 x 2-1/2 x 5	1
EE1-801	1	1	3		4100	3300	2400	3/4 x 2-3/4 x 5-1/2	1
EE1-801	1	1	4		5500	4500	3200	1 x 3-1/2 x 7	1
EE2-801	1	2	1	3000	–	–	–	1/2 x 2-1/2 x 5	1-1/2
EE2-801	1	2	2		5100	4200	3000	3/4 x 2-3/4 x 5-1/2	1-1/2
EE2-801	1	2	3		7700	6300	4500	3/4 x 2-3/4 x 5-1/2	1-1/2
EE2-801	1	2	4		10300	8400	6000	1 x 3-1/2 x 7	1-1/2
EE1-802	2	1	1	3000	–	–	–	1/2 x 2-1/2 x 5	1-1/2
EE1-802	2	1	2		5100	4200	3000	3/4 x 2-3/4 x 5-1/2	1-1/2
EE1-802	2	1	3		7700	6300	4500	3/4 x 2-3/4 x 5-1/2	1-1/2
EE1-802	2	1	4		10300	8400	6000	1 x 3-1/2 x 7	1-1/2
EE2-802	2	2	1	6000	–	–	–	3/4 x 2-3/4 x 5-1/2	3
EE2-802	2	2	2		10300	8400	6000	1 x 3-1/2 x 7	3
EE2-802	2	2	3		15500	12700	9000	1 x 3-1/2 x 7	3
EE2-802	2	2	4		20700	16900	12000	1-1/4 x 4-3/8 x 8-3/4	3

- SOE = Single leg with oblong master link and eye on bottom
- SOS = Single leg with oblong master link and sling hook on bottom

There are many options for end fittings. Please see Core Synthetic Sling Hardware Catalog for specifications.



WEB SLING SAFETY BULLETIN

WARNING

Lifting & lifting systems require the use of several tools, variables & devices, and polyester web sling(s) are only one part of that system. It is critical that you read and understand this bulletin prior to using polyester web slings and the failure to do so, may result in severe **INJURY** or **DEATH** due to sling failure and/or loss of the load.

The 6 Areas of Concentration

1. **TRAINING & KNOWLEDGE**
2. **GOOD INSPECTION PRACTICES**
3. **SLING PROTECTION**
4. **NEVER EXCEED RATED CAPACITY**
5. **STAY CLEAR OF LOAD**
6. **SLING STORAGE**

1. TRAINING & KNOWLEDGE

Knowledge is acquired through training. Sling users must be knowledgeable in the safe and proper use of polyester web slings to prevent **injury** and **death** to you and others.

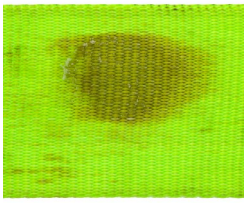
In addition to the information found in this bulletin, polyester web sling users should also read and understand the standards and guidance provided in:

- ASME B30.9-5 Synthetic Webbing Slings: Selection, Use & Maintenance
- OSHA 29 CFR 1910.184: Slings
- OSHA Guidance On Safe Sling Use

2. GOOD INSPECTION PRACTICES

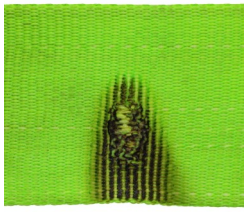
Because polyester web slings are made from synthetic fibers, they are vulnerable to damage in a variety of ways. Before using a sling to make a lift, it is necessary to inspect the sling for damage prior to use to prevent **injury** or **death** to you and others. Inspection requires you to **LOOK** & **FEEL** over the entire length of the sling on both sides.

During inspection, should you identify any of the following conditions, the sling is to be **removed from service** in accordance with ASME B30.9-5.9.4.



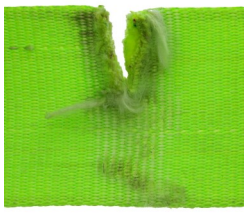
Acid or Alkali Burns

Melting or charring of any part of the sling

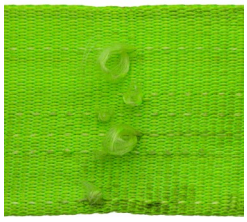


Holes/Punctures

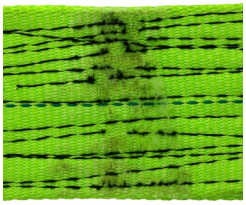
Cuts or Tears



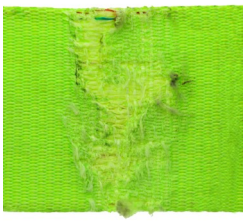
Snags



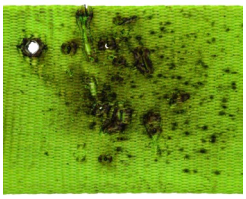
Knots



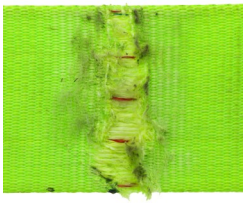
Broken or Worn Stitches



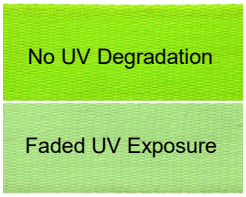
Excessive Abrasive Wear



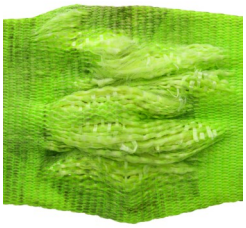
Weld Spatter



Exposed Red Core Yarns



UV Degradation



Crushed Webbing

Embedded Materials

Other removal criteria:

- Slings with fittings that are pitted, corroded, cracked, bent, twisted, gouged or broken.
- For slings with hooks sewn-in, apply removal criteria for hooks as stated in ASME B30.10.
- For other sewn-in rigging hardware, apply removal criteria as stated ASME B30.26.
- Other conditions, including visible damage, that cause doubt as to the continued use of the sling.

Polyester web slings should be inspected as follows:

Intially—When slings are received by a qualified person to ensure there is no damage and that the sling meets the requirements for its intended use.

Frequently—Polyester web slings should be inspected before using each shift or each day.

Periodically—Polyester web slings are to be inspected by a qualified person, other than the person who conducts the frequent inspections. Periodic inspections are to be done annually for normal service and monthly to quarterly for severe service. Inspection intervals are not to exceed one year. For periodic inspections, both WSTDA WS-1 and ASME B30.9 require that a written record of the inspections be maintained.

3. SLING PROTECTION

To protect polyester web slings from both environmental and physical damage, avoid the following:

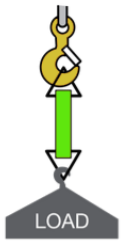
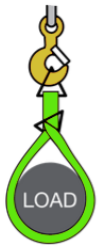
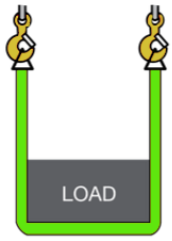
- Continuous or excessive exposure to sunlight.
- Contact with dirt or gritty type substances.
- Dragging slings on the floor, ground or rough surfaces.
- Jerking or pulling slings out from under loads.
- Twisting or tying knots in slings.
- Allowing slings to contact acids or alkalis.
- Exposing slings to temperatures above 194°F or below –40°F.
- Allowing weld spatter to land on slings.
- Allowing sling to sit on the tip of a hook instead of the base of the hook's bowl.
- Using rigging hardware such as hooks or shackles with rough edges.
- Running or driving over slings with vehicles or mobile equipment.

To help prevent damage from sharp edges, corners and other abrasive surfaces, protective devices such as wear pads, sleeves, corner protectors and body wraps should be used. It may also be necessary to test samples to ensure protection method chosen is effective prior to lifting actual load to be lifted.

4. NEVER EXCEED RATED CAPACITY

To avoid exceeding the rated capacity of a polyester web sling, a trained and qualified person must first know the weight of the load to be lifted and its center of gravity to determine if the rigging system will sufficiently retain and control the lifted load.

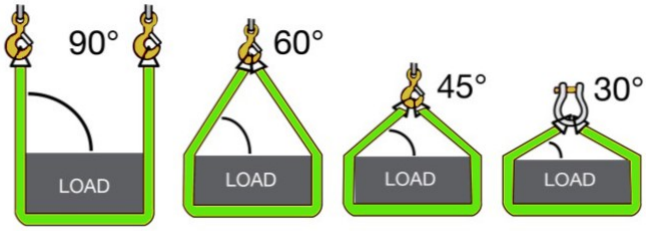
Also, determine whether the sling will be used in a vertical hitch, choker hitch or vertical basket hitch. See table below for description and illustration.

Vertical	Choker	Vertical Basket
		
One end is placed on the hook and the other end is attached directly to the load. A tagline should be used to prevent load rotation.	Sling passes thru one end & around the load while other end is placed on hook. Rated capacity is normally 80% of vertical hitch. Choke point should always be on sling body, not sling eye, fitting, splice or tag.	Sling cradles load while ends are attached overhead. Rated capacity is two times that of vertical hitch.

It is also very important that a qualified person determine the angle that the slings will be placed in for the intended lift. This may even be a challenge for a qualified person. But for someone unqualified and unaware of the effects that sling angles can have on the rated capacity of the slings, the lack of this knowledge or inability to apply the correct sling reduction factors can lead to injury or death.

WARNING: *ANGLES MATTER!*

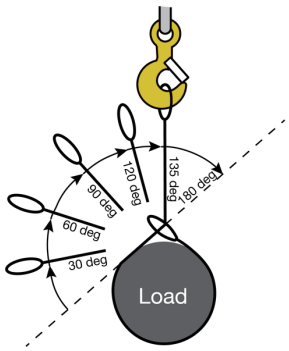
For Vertical & Basket hitches, sling angles less than *90 degrees* **reduce the effective rated capacity of the sling**. See table below.



Angle	Sling Reduction Factor
90°	1.000
85°	0.996
80°	0.985
75°	0.966
70°	0.940
65°	0.906
60°	0.866
55°	0.819
50°	0.766
45°	0.707
40°	0.643
35°	0.574
30°	0.500

Vertical & Basket Hitches

To calculate the effective capacity of the sling, determine the angle of the sling and multiply the sling reduction factor for the determined angle times the rated Vertical or Basket capacity of the sling.



Angle of Choke		Sling Reduction
= or >	<	Factor
120°	180°	1.00
105°	120°	0.82
90°	105°	0.71
60°	90°	0.58
0°	60°	0.50

For Choker hitches, to calculate the effective capacity of the sling, determine the angle of choke and multiply the sling reduction factor for the determined angle of choke times the rated Choker capacity of the sling.

5. STAY CLEAR OF LOAD

- Never stand in the area directly beneath or near the load being lifted.
- Never stand on load being lifted.
- Never allow any part of your body to be placed between the sling and the load being lifted or between the sling and the lifting hook or rigging fitting.
- Be sure all personnel in the area of the lift are alert to the potential for the sling to become snagged during the lift.
- Never use a web sling to pull on objects in a constrained or snagged condition.

6. SLING STORAGE

- Polyester web slings should always be stored in dry, dark and cool places when not being used.
- Polyester web slings should always be stored away from exposure to chemicals, high heat sources, metal turnings from machining, grinding dust, splinters, etc.
- Always be sure polyester web slings are stored in a place free from dirt, oils, grime and other such materials.
- Clean only with mild soap and water. If washed, be sure polyester web sling is 100% dry before being stored. Never machine wash as it will result in loss in strength.

There are many factors that should be considered when using web slings to lift loads. Some of these factors include:

- Wind
- Visibility
- Air Temperature
- Center of Gravity
- Clear Load Path
- Sling Selection
- Lift Point
- Signals
- Weather
- Ground Stability
- Weight
- Dimensions
- Structural Stability
- Load Control
- Sling Capacity
- Planning

The information in this bulletin does not contain ALL of the information required to use web slings safely. Other recommended sources of information include:

- OSHA 29CFR 1910.184
- Web Sling Tie Down Association (WSTDA) “WS-1” Recommended Standard for Synthetic Web Slings
- ASME B30.9 Sling Standard

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